DETONATOR, FLASH, XM97
PARTS FOR, AND LOADING
ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of flash detonator designated Detonator, Flash, XM97.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-14137 - Primer, Stab, M106 Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9254812 - Detonator, Flash, XM97 Assembly

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET 9254812 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, 25, D.C. 20402. Orders shall specify 49 CFR 0-190 (latest revision).)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The detonator assembly shall comply with all requirements specified on Drawing (dwg.) 9254812 and with all requirements specified in applicable specifications.

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3.3 Moisture content

3.3.1 Lead azide.- The moisture content of the lead azide at the loading station at the time of loading the detonator shall not exceed 0.50 percent when tested as specified in 4.4.1.1.

3.3.2 Pentaerythrite tetranitrate (PETN).- The moisture content of the PETN at the loading station at the time of loading the detonator shall not exceed 0.50 percent when tested as specified in 4.4.1.2.

3.4 Functioning.- The detonator assembly shall not fail to fire on the first attempt (dud) and shall make a 1/4 inch minimum (min.) diameter perforation in the lead disc when tested as specified in 4.4.2.

3.5 Waterproofness.- The detonator assembly shall comply with the functioning requirement specified in 3.4 when tested as specified in 4.4.3.

3.6 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, scale, dirt, grease, corrosion products, salt deposits and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.7 First article inspection.- This specification makes provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that statements of findings on the moisture content of the lead azide and PETN shall be furnished with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation as specified herein. The first article sample shall consist of inert parts (see 4.2.2) and loaded detonators (see 4.2.3) which have been

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produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inert parts.- A sample of fifty (50) each of every component shall be submitted in accordance with the provisions of 4.2.1 for inspection as specified in 4.2.2.1.

4.2.2.1 Inspections to be performed.- Components will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.2.2 Rejection.- If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an item to comply with any of the requirements.

4.2.3 Loaded detonators.- A sample of one hundred and twenty-five (125) loaded detonator assemblies shall be submitted in accordance with the provisions of 4.2.1 for testing as specified in 4.2.3.1.

4.2.3.1 Tests to be performed.- The loaded detonators will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specifications and any or all requirements of the applicable drawings.

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4.2.3.2 Rejection.- The loaded detonator first article sample shall be rejected if one or more detonators fail to comply with an examination or test requirement. The Government reserves the right to terminate the testing upon any failure of a detonator to comply with the requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded assemblies shall contain:

a. Parts of one lot interfix number from one supplier.
b. PETN from not more than one lot.
c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

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4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 9252743 covering a detail of dwg. 9254812)

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Crack or split	Visual	01006
Major	AQL 0.65 percent		
101.	Thickness through bottom, min.	Gage	01001
102.	Outside diameter, maximum (max.)	Gage	01002
103.	Dent or distortion	Visual	01003
Minor	AQL 1.00 percent		
201.	Inside diameter, min.	Gage	01004
202.	Evidence of poor workmanship other than critical above (see 3.6)	Visual	01005

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4.3.2.2 Plug (see dwg. 9254814 covering a detail of dwg. 9254812)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 percent		
101.	Large outside diameter min.	Gage	02001
102.	Small outside diameter	Gage	02002
103.	Depth of cavity	Gage	02003
104.	Diameter of cavity	Gage	02004
105.	Total length	Gage	02005
106.	Concentricity of cavity with outside diameter	Gage	02006
107.	Crack	Visual	02007
Minor	AQL 1.00 percent		
201.	Chamfer over max.	Visual	02008
202.	Evidence of poor workmanship other than crack (see 3.6)	Visual	02009

4.3.2.3 Plug loading assembly (see dwg. 9254813 covering a detail of dwg. 9254812)

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Explosive on exterior	Visual	03001
Major	AQL 0.40 percent		
101.	Charge missing or incomplete	Visual	03002
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	03003

4.3.2.4 Assembly (see dwg. 9254812)

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Cup cracked or split	Visual	04001

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Major	AQL 0.65 percent		
101.	Total length	Gage	04002
102.	Diameter at plug end	Gage	04003
103.	Crimp not full 360 degrees	Visual	04004
104.	Varnish seal missing or incomplete	Visual	04005
Minor	AQL 1.50 percent		
201.	Diameter of cup base, max.	Gage	04006
202.	Plug crack	Visual	04007
203.	Evidence of poor workmanship other than cup cracked or split (see 3.6)	Visual	04008

4.3.3 Testing

4.3.3.1 Moisture content (see 3.3) - Major defects

Table I

Explosive	Code No.
Lead azide	08001
PETN	08002

The contractor shall provide adequate controls to insure that the explosives comply with the requirements and shall test for verification a minimum of one sample of each explosive selected at the loading station at the time of loading from each eight hours production of detonator assemblies. A composite sample shall not be used. Failure of a sample to comply with the requirements shall cause the explosive represented by the sample, including any detonators loaded with the non-conforming explosive to be rejected. The contractor shall maintain adequate records to provide identification of the explosive used in the loaded detonator assemblies. Tests shall be performed as specified in 4.4.1.

4.3.3.2 Functioning (see 3.4) -Major defect.- Code No. 04009 - The sample size, acceptance or rejection of the lot shall be in accordance with Table II. The test shall be performed as specified in 4.4.2 using equipment as specified in 4.3.4. If more than one loading machine is used in the production of the inspection lot, the sample shall be selected so as to represent proportional quantities of production from each machine.

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Table II

Lot Size	Sample	Accept.	Reject
To 150,000	First 315	0	3
	Second 315	3	4
150,001 to 300,000	First 500	1	4
	Second 500	4	5
300,001 to 500,000	First 800	2	5
	Second 800	6	7
500,001 and over	First 1,250	3	7
	Second 1,250	8	9

4.3.3.3 Waterproofness (see 3.5) - Major defect - Code No. 04010.- Sampling shall be in accordance with MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.3.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 9254812 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Lead azide.- The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.4.1.2 PETN.- The moisture content of the PETN shall be determined as specified in 4.4.1.1 except that the drying time shall be for one hour.

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4.4.2 Functioning and waterproofness.- The detonator assemblies shall be assembled in the fixture specified in 4.3.4 and initiated by the flash of an M106 Primer. The M106 Primer shall be initiated as specified in MIL-P-14137. Any detonator that fails to fire or fails to make the required hole size perforation in the lead disc shall be classed defective.

4.4.3 Waterproofness.- The detonator assemblies shall be conditioned for forty-eight (48) hours min. at a two (2) to three (3) inch depth, in water maintained at seventy (70) plus or minus ten (10) degrees Fahrenheit (F). After conditioning, the detonator assemblies shall be functioned in accordance with the procedure specified in 4.4.2. Any detonator that fails to fire or make the required hole size perforation in the lead disc shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Level C.- The detonators shall be preserved and packed in conformance with Code of Federal Regulations CFR 49 Parts 0-190.

5.2 Marking.- The container marking shall specify quantity, nomenclature, manufacturer's lot number, date loaded and Department of Transportation marking for the particular explosive item.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

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6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, one copy of all ammunition data and data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-ND5
(2) SMUPA-DC

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6.5 Check test for possible deterioration.- The following test is designed to prevent detonators that have deteriorated being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

6.5.1 Functioning and waterproofness.- A sample of three hundred and fifteen (315) detonators shall be selected from each lot and tested as specified in 4.4.2. If three (3) or more detonators fail to comply with the requirements, the lot shall be rejected. If one or two defectives are found, a second sample of 315 detonators shall be selected and tested. If the total number of defectives from both samples equals or exceeds five (5), the lot shall be rejected.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1375-A-0114

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